



Motivation & aim

Job ads can inform education, career decisions and workforce planning. Inconsistent terminology, different formats and missing fields make them hard to analyze. *Mihna* structures postings into evidence of demand that students, employers and policymakers can explore through dashboards and bilingual questions.

Data collection & preparation

Custom scraping pipeline
JobSpy collection

Dashboard snapshot coverage

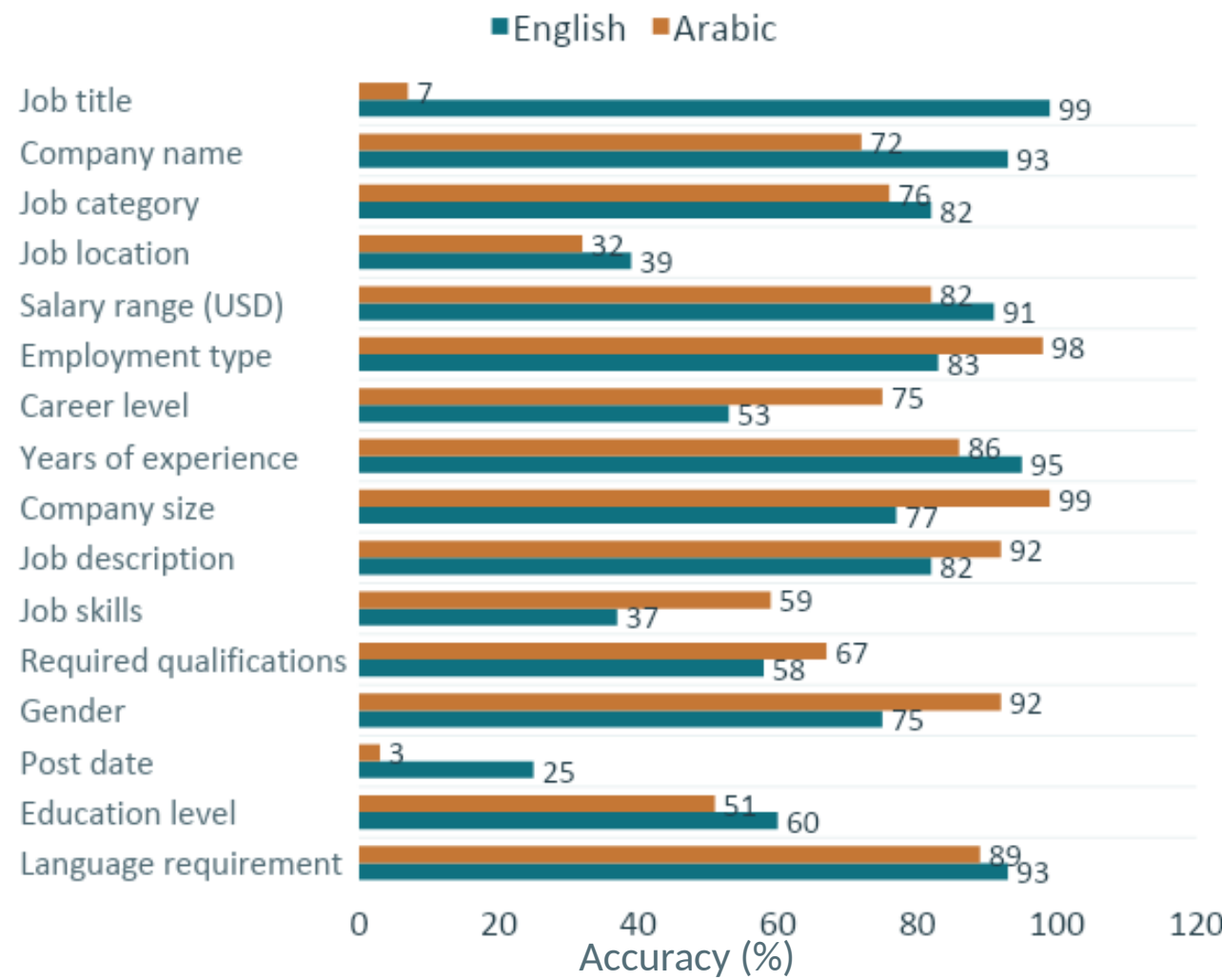
Source / snapshot	Qatar	Saudi Arabia	UAE
Bayt	10,131	16,101	19,124
LinkedIn	2,041	4,441	3,778

55,616 records across 11 dated datasets

Standardized metadata and remove duplicates. Used an LLM with a structured output schema to extract fields from job descriptions.

Extraction accuracy by field

GPT-5.1 vs human annotations · Bayt
150 English + 150 Arabic postings

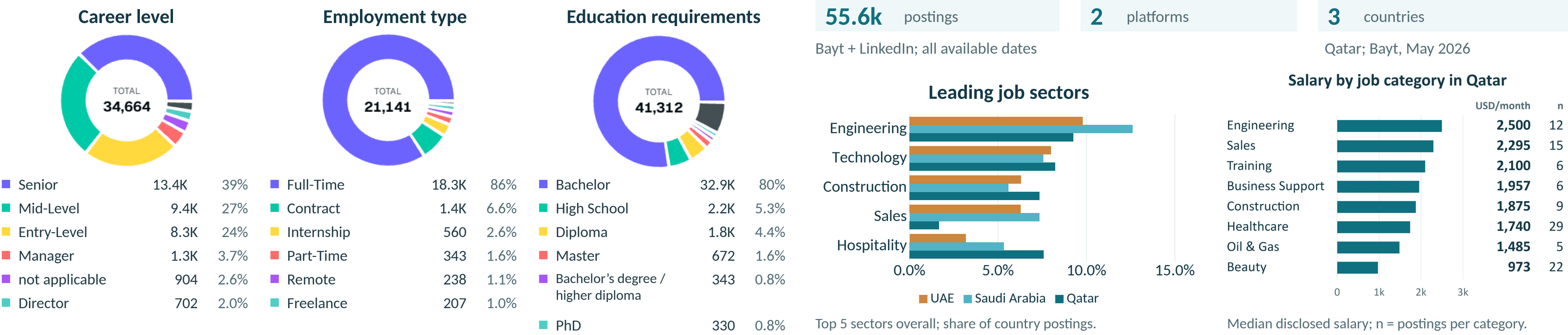


References

[1] Nguyen et al. (2024). Rethinking Skill Extraction in the Job Market Domain using Large Language Models. NLP4HR.
[2] Myung et al. (2025). HyST: LLM-Powered Hybrid Retrieval over Semi-Structured Tabular Data. arXiv:2508.18048.

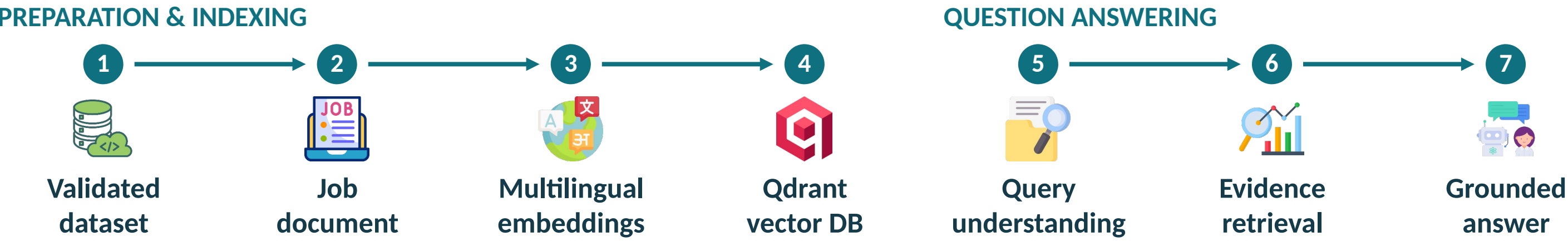


Dashboard exploration: Per-country and aggregated



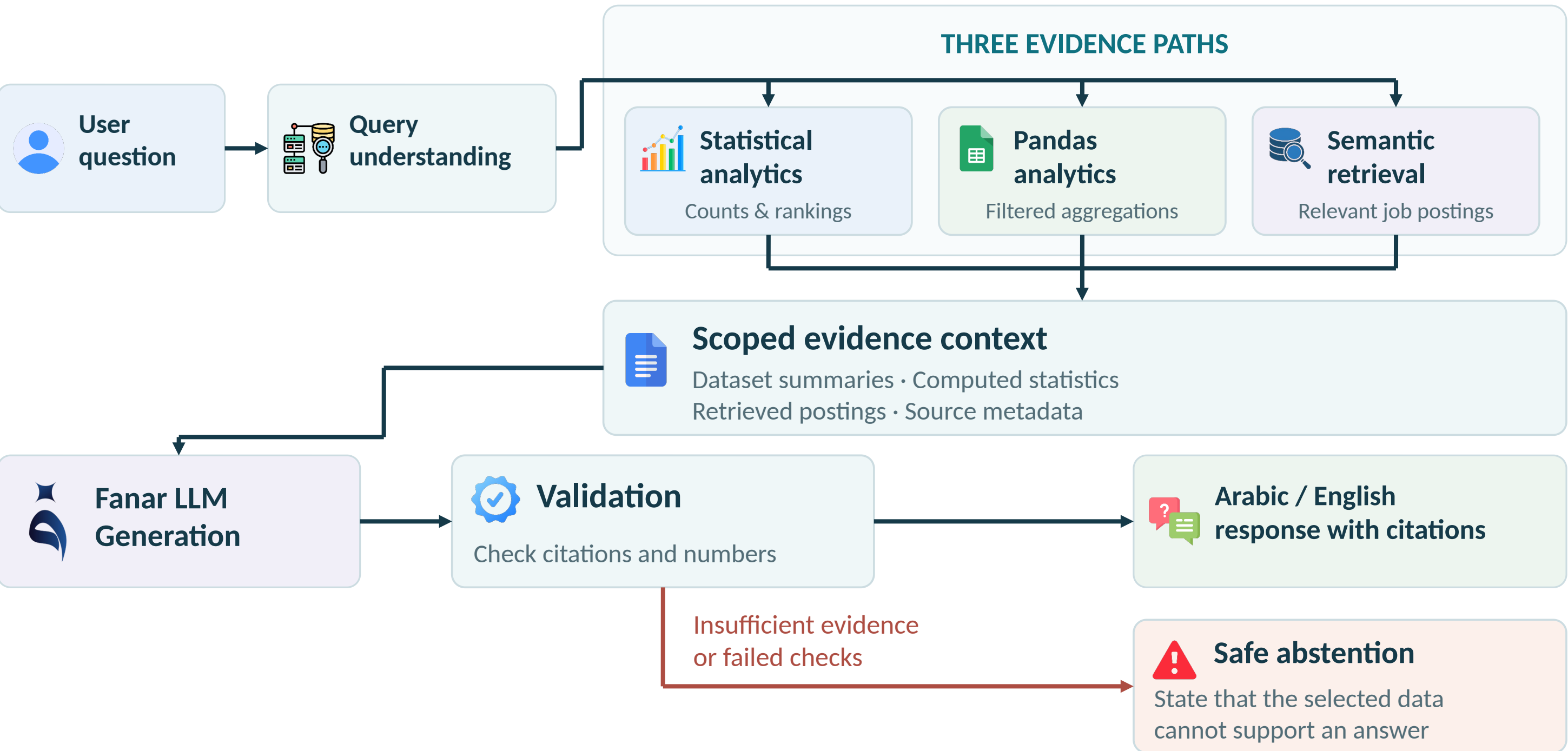
Bilingual RAG methodology

System overview



Online query answering

Retrieval and analytics are restricted to the selected source, country, period and dashboard filters.



Experiments & results

Benchmark: Automated scenario evaluation
26 English + 26 Arabic

51/52 Scenario checks
34/34 Evidence availability
52/52 Output language checks
3/4 Safe abstention

Platform conversation · Cropped excerpt

Which skills appear most often in data roles in Qatar?

Most requested skills in data roles

Based on 16 matching postings:

1. **Data Analysis** — 9 postings (56.2%)
2. **SQL** — 7 postings (43.8%)
3. **Communication** — 6 postings (37.5%)

...
Scope: Qatar · May 2026 · 2,376 postings

Future work

- Automate monthly collection, validation, cleaning and dashboard/index refreshes.
- Expand coverage across job sources, countries and comparable time periods.
- Explore demand forecasting and market reports as longitudinal coverage improves.